

Riley Brown

rydb@proton.me | rbrown71@nebraska.edu | ry3b.github.io | github.com/ry3b

EDUCATION

University of Nebraska-Lincoln

B.S. Mathematics and Data Science, Minor in Computer Science, Lincoln, NE | Aug 2023 - May 2027 (expected)

- GPA 3.92/4.00, University Honors Program
- President of Pi Mu Epsilon and Math Club; Putnam Exam Practice

PREPRINTS

Fixed-Parameter Tractability of t -Uniform Hypergraphicality

Riley Brown, Istvan Miklos

arXiv:2606.08523, 2026

RESEARCH EXPERIENCE

Undergraduate Researcher - Quantum LDPC Codes

University of Nebraska-Lincoln, Lincoln, NE | Fall 2026

- Testing whether collision-free geodesic-flow operators on Ramanujan complexes can sharpen iterative-decoder thresholds for quantum LDPC codes

Research Assistant, pseudorandomness

North Carolina State University, Raleigh, NC | Summer 2026

- Studied Ta-Shma's epsilon-balanced code construction with Prof. Chin Ho Lee as a research partner
- Found simpler constructions and minor improvements to the Hsieh-Mohanty-Zheng construction

Research Assistant, t -uniform hypergraph degree sequences and FPT algorithms

Budapest Semesters in Mathematics, Budapest, Hungary | Spring 2026

- Proved t -uniform hypergraphicality is FPT in (k, t) for compressed degree sequences, extending known bounded-range results
- Reduced realizability to integer programming feasibility via bounded-dimension spectra and balancing hinge-flips

Summer Researcher, algorithm design

Harvey Mudd College, Claremont, CA | Summer 2025

- Improved exponential-time algorithms for sumset recognition, an NP-complete problem in additive combinatorics (team of four)
- Sharpened branching strategies and base-case parameters for graph-theoretic reductions

Undergraduate Researcher, tiling theory

University of Nebraska-Lincoln, Lincoln, NE | Summer 2024

PROJECTS

leanverify

Lean 4 proof checker against Mathlib: rejects sorries and smuggled axioms | python, lean 4 | 2026

Deep learning for patient-safety monitoring

Parameter tuning and data analysis for Ocuvra to classify unattended hospital bed exits | python, deep learning

SKILLS

Programming: Python, MATLAB, Java, SQL, LaTeX, Lean 4

RELEVANT COURSEWORK

Bijjective Partition Theory, Discrete Mathematics, Matroid Theory, Advanced Combinatorics, Algebra I, Coding Theory, Development and Analysis of Algorithms, Graph Theory, Probability Theory, Automata, Computation, and Language, Data Structures and Algorithms